Strong and weak scaling

- **Strong scaling:** how solution time varies with the number of cores for a fixed **total** problem size.
  - Use 2X machines for a task => solve it in half the time
- **Weak scaling:** how solution time varies with the number of cores for a fixed problem size per cores.
  - if the dataset is twice as big, use 2X machines to solve the task in constant time.

Strong Scaling Speedup

- Strong scaling speedup = t(1)/t(N)
- t(1) = time with 1 processor
- t(N) = time with N processors

- Strong scaling efficiency = **Speedup/N**

Strong Scaling example

For all processor-counts, problem size is fixed. Compute speedup and efficiency

#Processors N	time in seconds T
1	64.4
2	33.9
4	17.4
8	8.7
16	4.7
32	2.7
64	1.6
128	1.0
256	1.4
512	3.69
1024	4.7

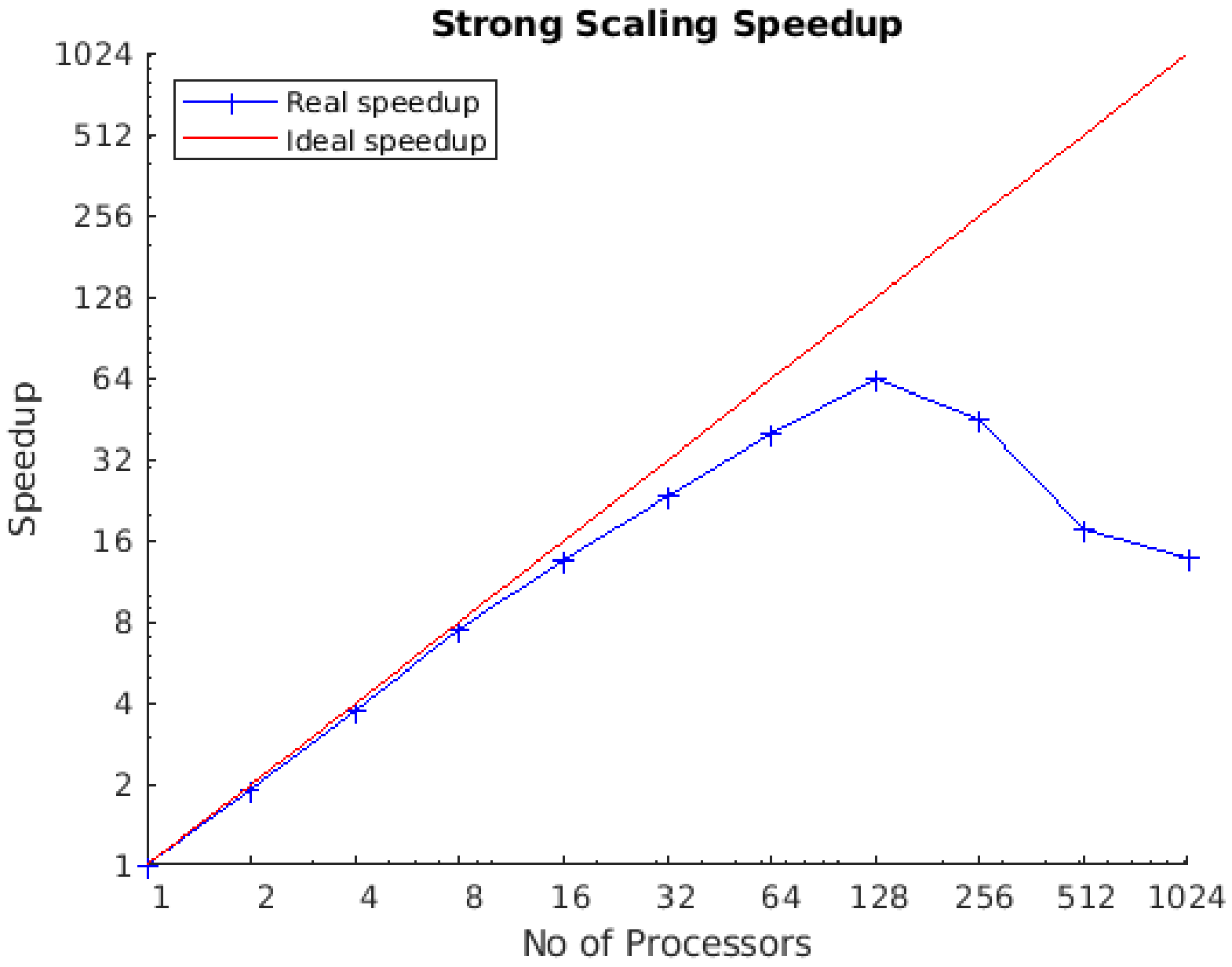
https://hpc-wiki.info/hpc/Scaling_tutorial

Strong Scaling example

For all processor-counts, problem size is fixed to 1600000000

#Processors N	time in seconds T	#speeedup = (T(1)/T(N))	Efficiency = Speedup/N
1	64.4	1	1
2	33.9	1.90	0.95
4	17.4	3.69	0.92
8	8.7	7.38	0.92
16	4.7	13.45	0.84
32	2.7	23.43	0.73
64	1.6	39.59	0.62
128	1.0	63.33	0.49
256	1.4	44.84	0.18
512	3.69	17.46	0.03
1024	4.7	13.68	0.01

https://hpc-wiki.info/hpc/Scaling_tutorial


Strong Scaling Speedup
Real speedup
Ideal speedup
Speedup
1024
512
256
128
64
32
16
8
4
2
1
1
2
4
8
16
32
64
128
256
512
1024
No of Processors

Weak Scaling Efficiency

- Amount of time to complete a work unit with 1 core is **t(1)**,
- Amount of time to complete N work units with N cores is **t(N)**
- Weak scaling efficiency is t(1)/t(N)

Weak Scaling example

Problem size is different for different processor-counts but problem size per core is fixed.

#Processors N	#problem size	time in seconds
1	16000000	0.58
2	32000000	0.63
4	64000000	0.72
8	128000000	0.78
16	256000000	0.89
32	512000000	0.97
64	1024000000	1.03
128	2048000000	1.26
256	4096000000	2.41
512	8192000000	4.59
1024	16384000000	10.69

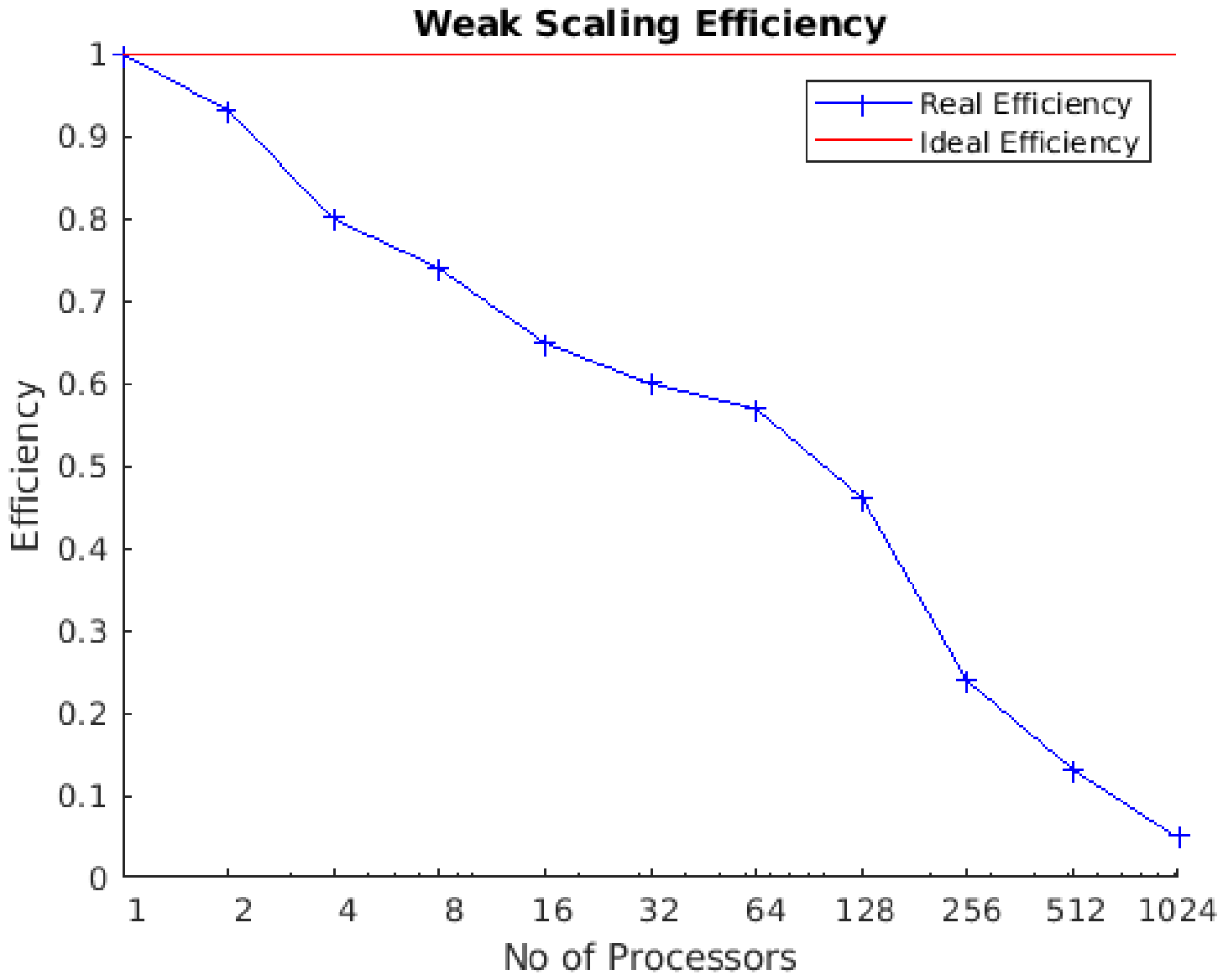
https://hpc-wiki.info/hpc/Scaling_tutorial

Weak Scaling example

Problem size is different for different processor-counts but problem size per core is fixed.

#Processors N	#problem size	time in seconds	Weak scaling efficiency = (T(1)/T(N))
1	16000000	0.58	1
2	32000000	0.63	0.93
4	64000000	0.72	0.80
8	128000000	0.78	0.74
16	256000000	0.89	0.65
32	512000000	0.97	0.60
64	1024000000	1.03	0.57
128	2048000000	1.26	0.46
256	4096000000	2.41	0.24
512	8192000000	4.59	0.13
1024	16384000000	10.69	0.05

https://hpc-wiki.info/hpc/Scaling_tutorial


Weak Scaling Efficiency
Real Efficiency
Ideal Efficiency
Efficiency
No of Processors
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
1
2
4
8
16
32
64
128
256
512
1024